Exact solution of the low-frequency coplanar loops shielding configuration

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Abstract: An analytical formulation is presented for the evaluation of the shielding effectiveness of a planar shield in the low-frequency, low-impedance antenna configuration described by IEEE Std. 299. The magnetic field computed in various shield configurations by means of the proposed closed-form expressions is presented. The finite size of both the transmitting and the receiving antenna is taken into account accurately. Finally, the exact solutions is compared with that deriving from the use of the so-called transmission line (TL) approximation, in various conditions.

1 Introduction

The configuration for the measurement of the shielding effectiveness (SE) of enclosures against low-frequency, low-impedance sources is described by IEEE Standard 299 [1] and its antecedent MIL-STD 285 [2], which prescribe that the two loop antennas (transmitting, TX, and receiving, RX) lie in a plane normal to the shield under test (coplanar loops). In the past, the exact solution for the SE has been found [3–5] only for the configuration with coaxial antennas, described in NSA 65-6 [6]. Even though formulations have been presented in the past [7, 8] for the shielding effectiveness in the coplanar loops configuration, the finite size of the transmitting and receiving loops has been rigorously accounted for the first time in [9], under the assumptions of constant current in the loop source and independency of the magnetic flux density on the position of the observation point on the shield surface. The effects of the finite size of the antennas are of practical interest: in fact, although the SE is introduced by definition as a local quantity, standards prescribe a measurement set-up which yields to results influenced by the finite size of the receiving loop. Moreover, the field illuminating the shield is also far from that due to a point source or a small (magnetic) dipole. In this paper the exact solution is thoroughly analysed and experimentally validated and the main hypotheses discussed.

It should be noted that in several textbooks and handbooks, tables and graphs are presented which assume without proof, that the small dipole approximation and the so-called Schelkunoff approach [10, 11] may be applied. Thus, in the configuration under analysis, applicability, limits and accuracy of predictions based on the Schelkunoff (or transmission line, TL) approach have been investigated, as well as errors due to the small dipole approximation.

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2 Exact solution

In order to solve the coplanar loops shielding problem, it is useful to introduce the magnetic vector potential of the second order, $\boldsymbol{U}$, as:

$$\nabla \times \boldsymbol{U} = \boldsymbol{A} \qquad \nabla \times (\nabla \times \boldsymbol{U}) = \boldsymbol{B} \tag{1}$$

where $\boldsymbol{A}$ and $\boldsymbol{B}$ are the magnetic vector potential and the magnetic flux density, respectively. Gauging $\boldsymbol{U}$ to have $\nabla \cdot \boldsymbol{U} = 0$, it follows that

$$\nabla^2 \boldsymbol{U} = 0 \tag{2}$$

Then, with reference to the geometrical configuration shown in Fig. 1, the vector components of the magnetic flux density are given by:

$$B_x = \frac{\partial^2 U_y}{\partial x \partial y} \tag{3a}$$

$$B_y = -\frac{\partial^2 U_y}{\partial x^2} - \frac{\partial^2 U_y}{\partial z^2} = \frac{\partial^2 U_y}{\partial y^2} \tag{3b}$$

$$B_z = \frac{\partial^2 U_y}{\partial y \partial z} \tag{3c}$$

where $\boldsymbol{U} = U_x \hat{\boldsymbol{x}} + U_y \hat{\boldsymbol{y}} + U_z \hat{\boldsymbol{z}}$. It should be noted that the magnetic flux density in air does not depend upon the x-

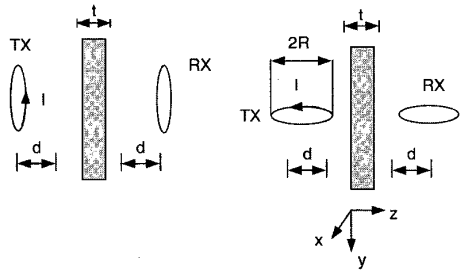


Fig. 1 *Comparison of set-up configuration for SE measurement according to NSA 65-6 and IEEE Std 299*
a NSA 65-6
b IEEE Std 299

and z-components of $\boldsymbol{U}$. An important simplification in the mathematical treatment is obtained by considering that the loop antenna is used up to 16 MHz and its diameter is $2R \cong 30.5$ cm; thus it can be assumed that

$$e^{-jkr} \cong 1 \tag{4}$$

where $k = \omega/c$ is the wavenumber and c is the speed of light in vacuum. Assuming also that the current in the source antenna does not depend on the angular position, it can be found [12] that in absence of the shield, the following expression for the magnetic vector potential of the second order holds:

$$\boldsymbol{U}_y(x, y, z) = \int_0^\infty \int_0^\infty G_0(\alpha, \beta)\cos(\alpha x)\cos(\beta y)e^{-D_1|z+d|}d\alpha\, d\beta \tag{5}$$

with

$$G_0 = -\frac{\mu_0 IR}{\pi}\frac{1}{\beta D_1}I_1(\beta R) \tag{6a}$$

$$D_1 = \sqrt{\alpha^2 + \beta^2} \tag{6b}$$

where μ_0 is the magnetic permeability of vacuum, R is the radius of the transmitting loop antenna, d is the minimum distance between the centre of the loop antenna and the shield, I is the RMS value of the sinusoidal current of angular frequency ω, and $I_1(.)$ is the modified Bessel function of the first kind of order one.

When the shield is applied, the magnetic vector potential of the second order is given by:

$$\boldsymbol{U}_y(x, y, z) = \int_0^\infty \int_0^\infty G_1(\alpha, \beta)\cos(\alpha x)\cos(\beta y)e^{-D_1|Z+d|}d\alpha\, d\beta \tag{7}$$

with

$$G_1 = -\frac{4\mu_0 IR}{\pi}\frac{D_3}{D_4}\frac{e^{-(D_2 t - D_1 t)}}{\beta}I_1(\beta R) \tag{8}$$

$$D_2 = \sqrt{\alpha^2 + \beta^2 + j\omega\mu\sigma} \tag{9a}$$

$$D_3 = \frac{D_2}{\mu_r} \tag{9b}$$

$$D_4 = (D_3 + D_1)^2 - (D_3 - D_1)^2 e^{-2D_2 t} \tag{9c}$$

where μ and σ are, respectively, the magnetic permeability and conductivity of the shield material, μ_r is the relative magnetic permeability, and t is the shield thickness.

Finally, it is possible to write an expression for the shielding effectiveness of the infinite shield under test, defined as

$$SE = 20\log_{10}\left|\frac{H_y^i}{H_y^t}\right| = 20\log_{10}\left|\frac{B_y^i}{B_y^t}\right| \tag{10}$$

where H_y^t, H_y^i, B_y^t and B_y^i are, respectively, the y-components of the magnetic field and magnetic flux density with and without the screen. Writing them out, with the expressions which can be readily obtained from eqns. 5 and 7, the SE is given by

$$SE = 20\log_{10}\left|\frac{-\int_0^\infty\int_0^\infty \beta^2 G_0(\alpha, \beta)\cos(\alpha x)\cos(\beta y)e^{-D_1|z+d|}d\alpha\, d\beta}{-\int_0^\infty\int_0^\infty \beta^2 G_1(\alpha, \beta)\cos(\alpha x)\cos(\beta y)e^{-D_1|z+d|}d\alpha\, d\beta}\right| \tag{11}$$

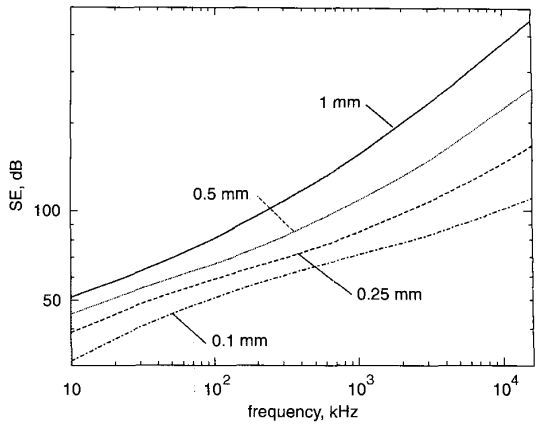


Fig. 2 *Theoretical SE against frequency calculated by exact theory for aluminium screen*

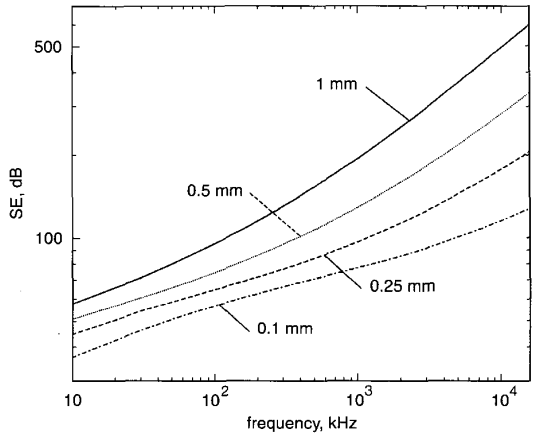


Fig. 3 *Theoretical SE against frequency calculated by exact theory for copper screen*

Even though this expression may appear complicated, the functions to be integrated are smooth and the computation of the double integrals is easy, e.g. by means of commercial packages or library routines. It should be noted that the shielding effectiveness is a function of the shield position between the two fixed loops, since it depends on both the distance d from the TX antenna and the distance z from the RX antenna. In Figs. 2–4, the shielding effectiveness at the centre of the receiving loop, evaluated in the range of frequency and with the set-up suggested by IEEE Std. 299 ($d = 3R$), is reported. Three materials have been selected for the analysis: the first having relative permeability (μ_r) and relative conductivity (σ_r) equal to one (annealed copper), the second having μ_r but not σ_r equal to one (household foil aluminum $\sigma_r = 0.53$), and the last having neither μ_r nor σ_r equal to unity (commercial iron 0.2% impure, $\mu_r = 200$, independent of frequency, $\sigma_r = 0.17$). Four different values have been considered for the thickness of each shield: 0.1, 0.25, 0.5 and 1 mm.

3 Considerations on experimental set-up

In order to compare the analytical predictions with measured data, it is important to account for the finite

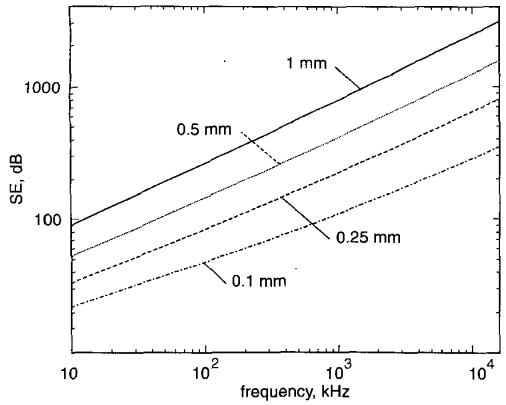


Fig. 4 *Theoretical SE against frequency calculated by exact theory for iron screen*

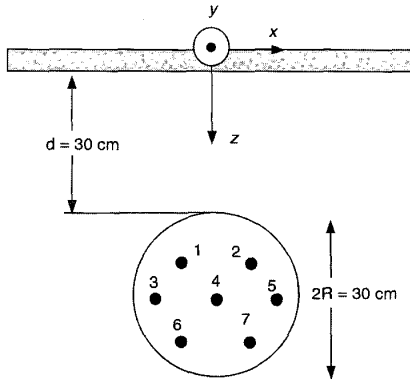


Fig. 5 *Location of seven points for approximate quadrature of double integral over circular surface*

dimensions of the receiving loop antenna. In fact, the component of the magnetic flux density normal to the RX antenna presents a spatial distribution which is dependent on the frequency and the shield characteristics. It should be noted that, when the voltage detected by the meter is proportional to the magnetic field intensity, Appendix B2 of IEEE Std. 299 suggests to compute the SE with the simpler expression

$$S = 20 \log_{10} \frac{V_{ns}^i}{V_s^t} = 20 \log_{10} \frac{\Phi_{ns}^i}{\Phi_s^t} \tag{12}$$

where V_{ns}^i and V_s^t are, respectively, the voltmeter readings in the absence and in the presence of the shield. However, the meter readings are proportional to the magnetic fluxes linked with the receiving loop, not to the magnetic field, and thus it is not possible to evaluate the field at a fixed position (e.g. the centre of the RX loop), assuming that values as a mean value. In order to avoid the computation of the double integral of the magnetic flux density over the surface described by the RX antenna, S_{RX}, it is possible to consider the following approximation [13]:

$$\iint_{S_{RX}} \boldsymbol{B}_y dS = S_{RX} \sum_{k=1}^{N_P} w_k \boldsymbol{B}_y(x_k, z_k) \tag{13}$$

In this way, the evaluation of the double integral is reduced to a much simpler evaluation of the magnetic flux density at some prefixed points. In particular, it has been found that a seven-point approximation ($N_P = 7$) allows a good accuracy. The points where the magnetic flux density has to be evaluated are shown in Fig. 5, and their coordinates and weights w_k in eqn. 13 are reported in Table 1. Due to the symmetry of the point locations and to eqn. 11, only four points need to be considered (e.g. 2, 4, 5 and 7). It should be noted that the systematic error ε on the SE at the centre of the RX loop, due to the use of eqn. 12 instead of eqn. 10, is evaluated as

$$\varepsilon = 100 \frac{B_y(0, 0, d + R) - 0.25 \sum_k B_k(P_k)}{B_y(0, 0, d + R)} \tag{14}$$

where $k = 2$, 4, 5 and 7, has been found to be practically independent of frequency, shield thickness and material characteristics. In Fig. 6, the SE of 0.25 mm thick aluminum, copper and iron screens evaluated by eqn. 10 is compared with the SE obtained by eqn. 12. The error has been found to be less than 5% over the total frequency range in all the cases analysed.

Table 1: Point coordinates and weights in eqn. 13

Point	x	Z	Weight
1	$-R/\sqrt{6}$	$d + R(1 - \sqrt{2}/2)$	0.125
2	$R/\sqrt{6}$	$d + R(1 - \sqrt{2}/2)$	0.125
3	$-R/\sqrt{2/3}$	$d+R$	0.125
4	0	$d+R$	0.25
5	$R/\sqrt{2/3}$	$d+R$	0.125
6	$-R/\sqrt{6}$	$d + R(1 + \sqrt{2}/2)$	0.125
7	$R/\sqrt{6}$	$d + R(1 + \sqrt{2}/2)$	0.125

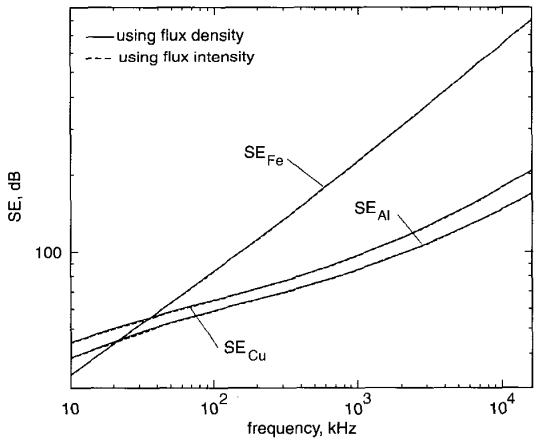


Fig. 6 *Theoretical SE against frequency calculated by exact theory using both magnetic flux density and magnetic flux intensity for aluminium, copper and iron screens of thickness 0.25 mm*

4 Suitability of TL approach

The transmission line (TL) analogy is often used to predict the shielding effectiveness of enclosures because of its simplicity. This approach is based on the following main assumptions:

(i) Maxwell's equations describing the electromagnetic field in the shield region have the same form as the TL equations;

(ii) the source can be considered ideal, that is the current is not influenced by the wave reflected by the screen;

(iii) the impedance of the incident wave, defined as the ratio between the tangential component of the electric and magnetic fields, is the same as the impedance of the transmitted one.

These conditions are rigorously verified for plane wave sources, and can be accepted in the coplanar configuration proposed by the IEEE-Std. 299 with some limitations and errors that have to be investigated. According to the expression given in Section 2, the electric and magnetic fields incident on the screen are:

$$E_x^{(1)}(x, y, z) = -j\omega\frac{\mu_0 IR}{\pi}\int_0^\infty\int_0^\infty \frac{1}{\beta} I_1(\beta R)\sin(\alpha x)\cos(\beta y)\mathrm{e}^{-D_1 z - D_1 d}\,d\alpha\,d\beta \tag{15a}$$

$$E_z^{(1)}(x, y, z) = -j\omega\frac{\mu_0 IR}{\pi}\int_0^\infty\int_0^\infty \frac{\alpha}{\beta D_1} I_1(\beta R)\sin(\alpha x)\cos(\beta y)\mathrm{e}^{-D_1 z - D_1 d}\,d\alpha\,d\beta \tag{15b}$$

$$H_x^{(1)}(x, y, z) = -\frac{\mu_0 IR}{\pi}\int_0^\infty\int_0^\infty \frac{\alpha}{D_1} I_1(\beta R)\sin(\alpha x)\sin(\beta y)\mathrm{e}^{-D_1 z - D_1 d}\,d\alpha\,d\beta \tag{15c}$$

$$H_y^{(1)}(x, y, z) = \frac{\mu_0 IR}{\pi}\int_0^\infty\int_0^\infty \frac{\beta}{D_1} I_1(\beta R)\cos(\alpha x)\cos(\beta y)\mathrm{e}^{-D_1 z - D_1 d}\,d\alpha\,d\beta \tag{15d}$$

$$H_z^{(1)}(x, y, z) = -\frac{\mu_0 IR}{\pi}\int_0^\infty\int_0^\infty I_1(\beta R)\cos(\alpha x)\sin(\beta y)\mathrm{e}^{-D_1 z - D_1 d}\,d\alpha\,d\beta \tag{15e}$$

The electric and magnetic fields of the first forward-travelling wave inside the shield are:

$$E_x^{(2)}(x, y, z) = -j\omega\frac{\mu_0 IR}{\pi}\int_0^\infty\int_0^\infty \frac{2\beta}{D_1(D_3 + D_1)} \times I_1(\beta R)\cos(\alpha x)\cos(\beta y)\mathrm{e}^{-D_2 z - D_1 d}\,d\alpha\,d\beta \tag{16a}$$

$$E_y^{(2)}(x, y, z) = -j\omega\frac{\mu_0 IR}{\pi}\int_0^\infty\int_0^\infty \frac{2\alpha}{D_1(D_3 + D_1)} \times I_1(\beta R)\sin(\alpha x)\sin(\beta y)\mathrm{e}^{-D_2 z - D_1 d}\,d\alpha\,d\beta \tag{16b}$$

$$H_x^{(2)}(x, y, z) = -\frac{IR}{\mu_r\pi}\int_0^\infty\int_0^\infty \frac{2\alpha D_2}{D_1(D_3 + D_1)} \times I_1(\beta R)\sin(\alpha x)\sin(\beta y)\mathrm{e}^{-D_1 z - D_1 d}\,d\alpha d\beta \tag{16c}$$

$$H_y^{(2)}(x, y, z) = -\frac{IR}{\mu_r\pi}\int_0^\infty\int_0^\infty \frac{2\beta D_2}{D_1(D_3 + D_1)} \times I_1(\beta R)\cos(\alpha x)\cos(\beta y)\mathrm{e}^{-D_1 z - D_1 d}\,d\alpha\,d\beta \tag{16d}$$

$$H_z^{(2)}(x, y, z) = -\frac{IR}{\mu_r\pi}\int_0^\infty\int_0^\infty \frac{2D_1}{D_3 + D_1} \times I_1(\beta R)\cos(\alpha x)\sin(\beta y)\mathrm{e}^{-D_1 z - D_1 d}\,d\alpha\,d\beta \tag{16e}$$

The electric and magnetic fields beyond the screen due to the transmission of the first wave travelling forward through the screen are:

$$E_x^{(3)}(x, y, z) = -j\omega\frac{\mu_0 IR}{\pi}\int_0^\infty\int_0^\infty \frac{4D_1 D_3}{\beta(D_1 + D_3)^2} \times I_1(\beta R)\,\mathrm{e}^{-D_2 t + D_1 t}\cos(\alpha x)\cos(\beta y)\,\mathrm{e}^{-D_1 z - D_1 d}\,d\alpha\,d\beta \tag{17a}$$

$$E_z^{(3)}(x, y, z) = -j\omega\frac{\mu_0 IR}{\pi}\int_0^\infty\int_0^\infty \frac{4\alpha D_3}{\beta(D_1 + D_3)^2} \times I_1(\beta R)\,\mathrm{e}^{-D_2 t + D_1 t}\sin(\alpha x)\cos(\beta y)\,\mathrm{e}^{-D_1 z - D_1 d}\,d\alpha\,d\beta \tag{17b}$$

$$H_x^{(3)}(x, y, z) = -\frac{\mu_0 IR}{\pi}\int_0^\infty\int_0^\infty \frac{4\alpha D_3}{(D_1 + D_3)^2} \times I_1(\beta R)\,\mathrm{e}^{-D_2 t + D_1 t}\sin(\alpha x)\sin(\beta y)\,\mathrm{e}^{-D_1 z - D_1 d}\,d\alpha\,d\beta \tag{17c}$$

$$H_y^{(3)}(x, y, z) = \frac{\mu_0 IR}{\pi}\int_0^\infty\int_0^\infty \frac{4\beta D_3}{(D_1 + D_3)^2} \times I_1(\beta R)\,\mathrm{e}^{-D_2 t + D_1 t}\cos(\alpha x)\cos(\beta y)\,\mathrm{e}^{-D_1 z - D_1 d}\,d\alpha\,d\beta \tag{17d}$$

$$H_z^{(3)}(x, y, z) = \frac{\mu_0 IR}{\pi}\int_0^\infty\int_0^\infty \frac{4D_1 D_3}{(D_1 + D_3)^2} \times I_1(\beta R)\,\mathrm{e}^{-D_2 t + D_1 t}\cos(\alpha x)\sin(\beta y)\,\mathrm{e}^{-D_1 z - D_1 d}\,d\alpha\,d\beta \tag{17e}$$

Previous expressions show that the TL approximation can be considered only along the $\hat{z}$-axis, where the propagation of a travelling electromagnetic wave constituted by E_x and H_y occurs. Moreover, in order to investigate the suitability of the TL approach, the wave impedance and the transmission coefficient at the interfaces must be evaluated. The wave impedance of the incident field at $z=0$ can be defined as

$$Z_{w1} = \frac{E_x^{(1)}}{H_y^{(1)}} = -j\omega\mu_0\frac{\int_0^\infty\int_0^\infty \frac{I_1(\beta R)}{\beta}\,\mathrm{e}^{-D_1 d}\,d\alpha\,d\beta}{\int_0^\infty\int_0^\infty \frac{\beta I_1(\beta R)}{D_1}\,\mathrm{e}^{-D_1 d}\,d\alpha\,d\beta} \tag{18}$$

while the wave impedance of the forward travelling wave beyond the screen at $z=t$ is

$$Z_{w2} = \frac{E_x^{(3)}}{H_y^{(3)}} = -j\omega\mu_0\frac{\int_0^\infty\int_0^\infty \frac{4D_1 D_3}{(D_1+D_3)^2\beta} I_1(\beta R)\mathrm{e}^{-D_1 d - D_2 t}\,d\alpha\,d\beta}{\int_0^\infty\int_0^\infty \frac{4D_3\beta}{(D_1+D_3)^2} I_1(\beta R)\mathrm{e}^{-D_1 d - D_2 t}\,d\alpha\,d\beta} \tag{19}$$

Fig. 7 shows that the impedances at the two interfaces are different, so that the third hypothesis, essential for the TL approach, is not satisfied. Nevertheless, it can be shown that the reflections at the two interfaces of the screen obey equations of approximately the same form as the equations of a transmission line. In fact, the transmission coefficient

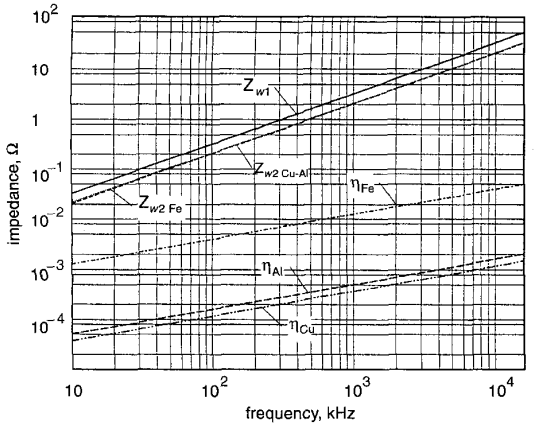


Fig. 7 *Theoretical wave impedances used in Schelkunoff approach*
Z_{w1} = incident wave impedance
Z_{w2} = transmitted wave impedance
η_{Al} = intrinsic wave impedance for aluminium screen
η_{Cu} = intrinsic wave impedance for copper screen
η_{Fe} = intrinsic wave impedance for iron screen

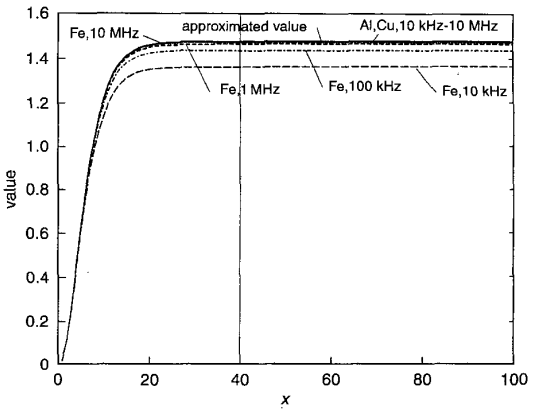


Fig. 8 *Trend of integral at numerator in eqn. 21, computed against x for aluminium, copper and iron screens compared with approximated value*

T_E at the first interface is written as

$$T_E = \frac{E_x^{(2)}}{E_x^{(1)}} = \frac{\int_0^\infty \int_0^\infty \frac{2\beta}{D_1(D_3+D_1)} I_1(\beta R) e^{-D_1 d} d\alpha\, d\beta}{\int_0^\infty \int_0^\infty \frac{1}{\beta} I_1(\beta R) e^{-D_1 d} d\alpha\, d\beta} \quad (20)$$

which can be set in the following convenient form:

$$T_E = \frac{2\eta}{Z_{w1}} \frac{\int_0^\infty \int_0^\infty \frac{1}{\mu_r} \frac{D_2}{D_3+D_1} \frac{\beta}{D_1} I_1(\beta R) e^{-D_1 d} d\alpha\, d\beta}{\int_0^\infty \int_0^\infty \frac{\beta}{D_1} I_1(\beta R) e^{-D_1 d} d\alpha\, d\beta} \quad (21)$$

where η is the wave impedance inside the screen at $z = 0$, given by

$$\eta = -j\omega\mu_0\mu_r \frac{\int_0^\infty \int_0^\infty \frac{2\beta}{D_1(D_3+D_1)} I_1(\beta R) e^{-D_1 d} d\alpha\, d\beta}{\int_0^\infty \int_0^\infty \frac{2\beta D_2}{D_1(D_3+D_1)} I_1(\beta R) e^{-D_1 d} d\alpha\, d\beta} \quad (22)$$

It may be easily verified that the above integrals converge quickly because of the negative exponential factor. Hence, there is an equal value of α and β, denoted $\bar{x}$, so that restricting the integration interval to $0 - \bar{x}$ in both the integrals does not affect the final value. In Fig. 8, the value of the integral at the numerator in eqn. 21 is shown against the upper limit of integration; a value of $\bar{x}$ of around 40 appears to be adequate for all the examined cases. In this range the factor $(1/\mu_r)[D_2(D_3 + D_1)]$ is, with excellent approximation, equal to one in the case of aluminum or copper shields, whereas in case of iron it ranges from 0.8 to 1 for increasing frequencies. Fig. 8 shows how this approximation is well suited in the aluminum and copper cases. Under these conditions, the transmission coefficient T_E is given by

$$T_E = \frac{2\eta}{Z_{w1}} \quad (23)$$

which is equal to the line transmission coefficient

$$T_E = \frac{2\eta}{Z_{w1} + \eta} \quad (24)$$

with the assumption that $Z_{w1} \gg \eta$, as is usual in practice.

Unfortunately, a completely different situation occurs in dealing with ferromagnetic materials; the assumption that the impedance of the incident wave is much larger than the intrinsic impedance of the shield is not correct as shown in Fig. 7. On the second interface, at $z = t$, the transmission factor is

$$T_E = \frac{E_x^{(3)}}{E_x^{(2)}} = \frac{\int_0^\infty \int_0^\infty 4 \frac{D_1 D_3}{(D_3+D_1)^2 \beta} I_1(\beta R) e^{-D_2 t - D_1 d} d\alpha\, d\beta}{\int_0^\infty \int_0^\infty 2 \frac{\beta}{D_1(D_3+D_1)} I_1(\beta R) e^{-D_2 t - D_1 d} d\alpha\, d\beta} \quad (25)$$

It is important to point out that Helmholtz's equation has been approximated by neglecting the displacement currents. For low frequencies, and in general for frequencies with wavelengths large compared with the thickness of the screen, this assumption is well suited, because it yields a negligible error, as will be shown in Section 5. In order to obtain the fields in the three regions bordered by the screen, the continuity of the normal component of the magnetic flux density and the continuity of the tangential components of the magnetic field intensity on the dividing surfaces were applied, whereas no conditions on the electric field components were imposed. If on the first interface, the difference between the $\hat{x}$-components of the electric field intensity in the vacuum and in the shield is so small that it can be neglected with good approximation; on the second one the component in the vacuum is much larger than the component in the shield, thus making impossible to justify the transmission line analogy using the electric field. Actually, the transmission coefficient quoted in eqn. 25, in all the examined cases, is around 3.8, which is completely different from the predictions carried out according to the TL analogy that would yield 2, since

$$T_E = 2\frac{Z_{w2}}{Z_{w2} + \eta} \cong 2 \quad (26)$$

with the assumption that $Z_{w2} \gg \eta$, as is evident from Fig. 7. Thus, considering that the problem examined here is to shield the magnetic field produced by a low-impedance loop source, it is convenient to estimate the transmission coefficient of the magnetic field, written as

$$T_M = \frac{H_y^{(3)}}{H_y^{(2)}} = \frac{\int_0^\infty \int_0^\infty \frac{4\beta D_3}{(D_3+D_1)^2} I_1(\beta R) e^{-D_2 t - D_1 d} d\alpha\, d\beta}{\int_0^\infty \int_0^\infty \frac{2\beta D_3}{D_1(D_3+D_1)} I_1(\beta R) e^{-D_2 t - D_1 d} d\alpha\, d\beta} \quad (27)$$

In the case of aluminum and copper, the value T_M ranges from 0.005 to 0.0001 for increasing frequencies in the range

10 kHz–16 MHz. This result agrees well with the expression of the TL approach:

$$T_M = 2\frac{\eta}{Z_{w2}+\eta} \cong 0 \tag{28}$$

when the usual approximation $Z_{w2} \gg \eta$ is introduced. As to ferromagnetic materials, the value of T_M ranges from 0.5 to 0.01 at frequencies of interest. The result at 10 kHz can be explained again considering that the intrinsic impedance of iron is around one order of magnitude smaller than Z_{w2}, as shown in Fig. 7, and thus the assumption $Z_{w2} \gg \eta$ is incorrect.

At the end of the analysis it is possible to conclude that, since the reflections at the screen interfaces and the fields inside the screen along the $\hat{z}$-axis are governed by equations of approximately the same form as the TL equations, within the approximations and restrictions pointed out, the TL approach can be properly applied in order to have a first assessment of the shielding effectiveness of a planar screen. However, as a final remark, it is worth mentioning that TL analogy is expected to be inaccurate in the case of ferromagnetic shields, often used in LF applications.

5 Analysis of approximations

Numerous approximations and assumptions are frequently introduced in the analysis of shielding problems, in order to simplify the analytical complexity and to make the results more manageable and easily applicable. The analysis of the assumptions seems to be useful in order to assess their range of validity. Herein, five kinds of error source are investigated:

(i) low-frequency antenna (eqn. 1): approximation as per eqn. 4;

(ii) low-frequency antenna (eqn. 2): constant current along the source antenna;

(iii) small dipole antenna;

(iv) transmission line theory, as explained in Section 4;

(v) same space dependence of the magnetic field in absence and in presence (beyond) of the shield.

The error due to the first approximation is a few per cent on the complex exponential and much less on the results of integrations, at the maximum frequency of interest.

As to the second assumption, it can be verified that $k_0.R \cong 0.5$ at the maximum frequency of 16 MHz, and thus, the current is practically constant (RMS value and phase) along the loop, as shown in [14, 15], with the error negligible.

In Figs. 9–11, the magnetic SE data evaluated by exact theory for the previous three types of materials of the shield are compared with those calculated by means of the small dipole approximation and the transmission line approach, where both the exact wave impedances and the small dipole approximated ones have been used. According to the theory explained in [7], the magnetic flux densities B_y^i and B_y^t produced by a y-directed magnetic dipole, in the absence and presence of the shield, have been respectively evaluated with the following expressions:

$$B_y^i = \frac{1}{8}\mu_0 k_0^3 IR^2 \int_0^\infty \left[\frac{\xi}{\zeta} - \xi\zeta\right] e^{-\zeta k_0(z+d)} d\xi \tag{29}$$

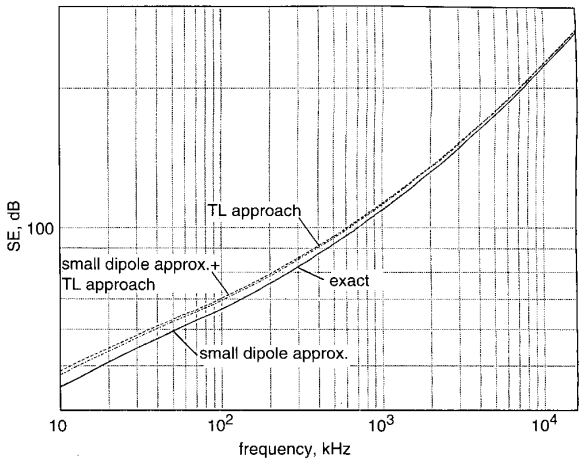


Fig. 9 *Theoretical SE against frequency calculated by exact theory, small dipole approximation and transmission line analogy using both exact wave impedances and small dipole impedances for aluminium screen of thickness 0.5 mm*

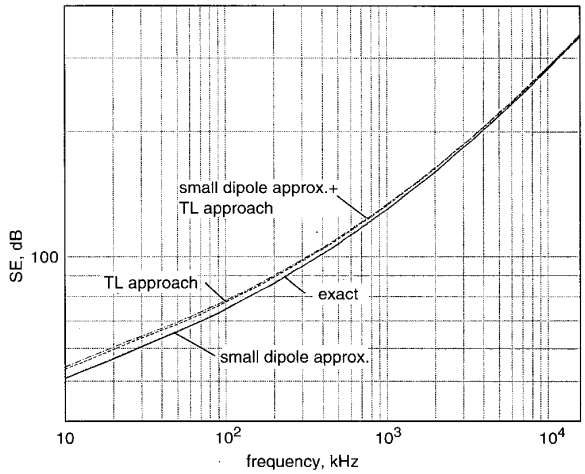


Fig. 10 *Theoretical SE against frequency calculated by exact theory, small dipole approximation and transmission line analogy using both exact wave impedances and small dipole impedances for copper screen of thickness 0.5 mm*

$$\begin{aligned} B_y^t &= \frac{1}{8}\mu_0 k_0^3 IR^2 \left(\int_0^\infty \frac{e^{(\zeta-\zeta_m)k_0 t}}{1 - \frac{(\varepsilon_r\zeta-\zeta_m)^2}{4\varepsilon_r\zeta\zeta_m}(e^{-2\zeta_m k_0 t} - 1)} \frac{\xi}{\zeta} e^{-\zeta k_0(z+d)} d\xi \right) \\ &= \frac{1}{8}\mu_0 k_0^3 IR^2 \left(\int_0^\infty \frac{e^{(\zeta-\zeta_m)k_0 t}}{1 - \frac{(\mu_r\zeta-\zeta_m)^2}{4\mu_r\zeta\zeta_m}(e^{-2\zeta_m k_0 t} - 1)} \xi\zeta e^{-\zeta k_0(z+d)} d\xi \right) \end{aligned} \tag{30}$$

where $\zeta = \sqrt{\xi^2 - 1}$ and $\zeta_m = \sqrt{\xi^2 - \mu_r\varepsilon_r}$. For a shield with conductivity σ, the relation $\varepsilon_r \cong (-j\sigma)/(\omega\varepsilon_0)$ holds. As concerns the TL approach, Schelkunoff's transmission theory [10] has been applied with the correct impedance Z_{w1} of the incident wave and the correct intrinsic impedance η within the shield, as given, respectively, by eqns. 18 and 22. Furthermore, the TL approach in conjunction with the small dipole approximation has been considered. In this

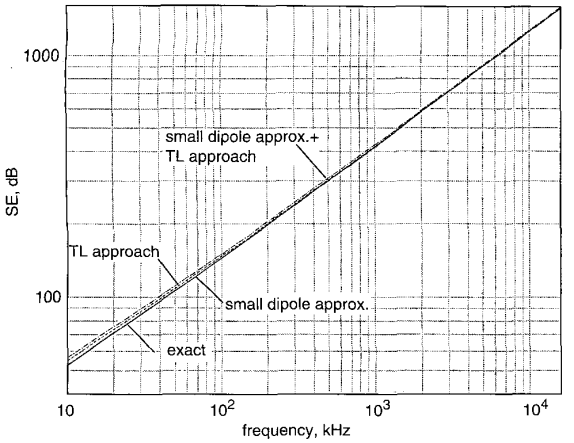


Fig. 11 *Theoretical SE against frequency calculated by exact theory, small dipole approximation and transmission line analogy using both exact wave impedances and small dipole impedances for iron screen of thickness 0.5 mm*

case the following expressions for the impedances have been used:

$$Z_{w1} = \eta_0 \frac{(j\omega)^2\left(\frac{d}{c}\right)^2 + j\omega\left(\frac{d}{c}\right)}{(j\omega)^2\left(\frac{d}{c}\right)^2 + j\omega\left(\frac{d}{c}\right) + 1} \tag{31}$$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} \tag{32}$$

From the previous Figures it should be noted the excellent agreement over the total frequency range between the exact theory and the small dipole approximation needs only a simple integration, which is rapidly performed following the hints given in [7]. The TL approach data, when the exact wave impedances are used, is always within 5 dB of the exact result, whereas the TL approach in conjunction with the small dipole approximation gives lower values. Fig. 12 shows the spatial variation of the component of the magnetic flux density, produced by a unit current source ($f = 14$ kHz), perpendicular to the receiving loop, in the area beyond the shield, along the line from one loop centre to the other, in the absence of the shield. The small dipole approximation with infinite terms and the usual one with three terms have also been applied. It is evident that if the former agrees well with the exact trend over the whole distance range, the latter introduces a large percentage error, which is shown in the zone of the RX antenna in Fig. 13.

Fig. 14 shows the trend of the component B_y radiated by a unit current source of $f = 14$ kHz beyond a shield of thickness 1 mm. Following the set-up proposed by IEEE Std. 299, the TX antenna has a diameter $2R \cong 30.5$ cm and its centre is located at $d = 45$ cm from the shield surface. The exact solution is compared with the trend obtained assuming a space-dependence of B_y as $1/r^3$. The figure clarifies that when the shield is applied, B_y does not have the same space dependence as in free space, so that care must be exercised in using the TL approach.

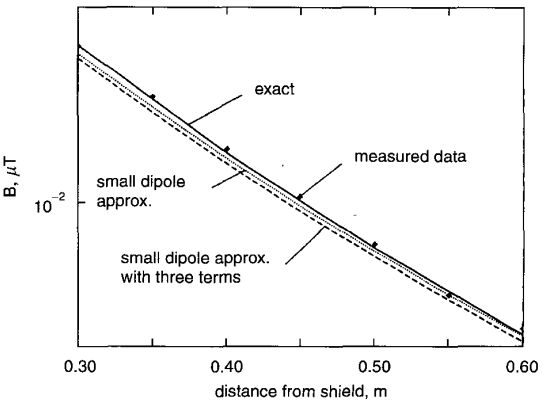


Fig. 12 *Trend of magnetic flux density in absence of shield ($f = 14$ kHz), calculated by exact theory and small dipole approximation with both infinite and three terms, compared with measured data*

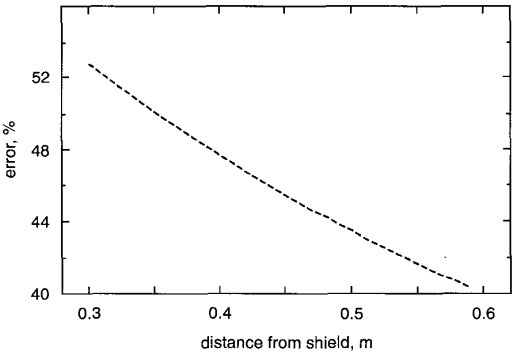


Fig. 13 *Percentage error due to small dipole approximation (three terms) as function of distance from shield (removed), in shielded region, in correspondence of RX antenna ($f = 14$ kHz)*

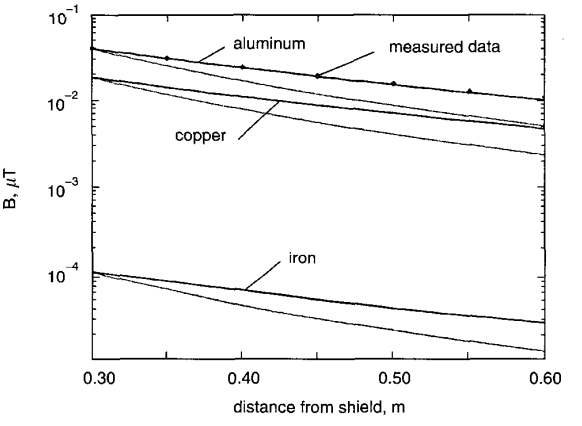


Fig. 14 *Trend of magnetic flux density in shielded region compared with $1/r^3$ law for aluminium, copper and iron screens ($t = 1$ mm, $f = 14$ kHz)*
Measured data for aluminium shield are also reported

6 Conclusions

Exact expressions and a numerical procedure have been presented for the analysis of low-frequency shielding in the low-impedance configuration described in IEEE Std. 299. The procedure allows one to evaluate the shielding effectiveness accurately and to have confidence in the approximations or the implicit assumptions usually adopted, such as the accuracy and the applicability of the TL approach or the correctness of the equivalence between SE definition and SE measurement setup.

7 References

1 MIL-STD 285: 'Method of military standard attenuation measurement for enclosures, electromagnetic shielding, for electronic test purposes', 1956
2 IEEE 299: 'IEEE standard method for measuring the effectiveness of electromagnetic shielding of enclosures', 1997
3 NSA: 65-5: 'National security agency specification for RF shielded enclosures for communications equipment: general specifications', 1964
4 MOSER, J.R.: 'Low-frequency shielding of a circular loop electromagnetic field source', *IEEE Trans. Electromagn. Compat.*, 1967, **EMC-9**, pp. 6–18
5 WHITEHOUSE, A.C.D.: 'Screening: new wave impedance for the transmission line analogy', *Proc. IEE*, 1969, **116**, (7), pp. 1159–1164
6 CELOZZI, S.: 'Shielding effectiveness prediction in low-frequency low-impedance source configurations'. Proceedings of 12th EMC Zurich symposium on *Electromagnetic compatibility*, Zurich, Switzerland , 1997, pp. 251–256
7 NISHIKATA, A., and SUGIURA, A.: 'Analysis for electromagnetic leakage through a plane shield with an arbitrarily-oriented dipole source', *IEEE Trans. Electromagn. Compat.*, 1992, **34**, pp. 284–291
8 YANG, R., and MITTRA, R.: 'Coupling between two arbitrarily oriented dipoles through multilayered shields', *IEEE Trans. Electromagn. Compat.*, 1985, **27**, pp. 131–136
9 CELOZZI, S.: 'Shielding effectiveness prediction in MIL-STD 285 loop source configuration'. Proceedings of IEEE 1998 international symposium on *Electromagnetic compatibility*, Denver, USA , 1998, pp. 1125–1130
10 SCHELKUNOFF, S.A.: 'Electromagnetic waves' (Van Nostrand, NY, USA, 1943)
11 SCHULZ, R.B., PLANTZ, V.C., and BRUSH, D.R.: 'Shielding theory and practice', *IEEE Trans. Electromagn. Compat.*, 1988, **30**, pp. 187–201
12 TEGOPOULOS, J.A., and KRIEZIS, E.E.: 'Eddy currents in linear conducting media' (Elsevier, Amsterdam, 1985)
13 ABRAMOWITZ, M., and STEGUN, I.A.: 'Handbook of mathematical functions' (Dover Publications Inc., New York, 1972)
14 BALANIS, A.C.: 'Antenna theory' (J. Wiley & Sons, NY, 1997)
15 STORER, J.E.: 'Impedance of thin wire loop antennas', *AIEE Trans.*, 1956, **75**, (Part I), pp. 606–619